Bilevel tender game: model formulation

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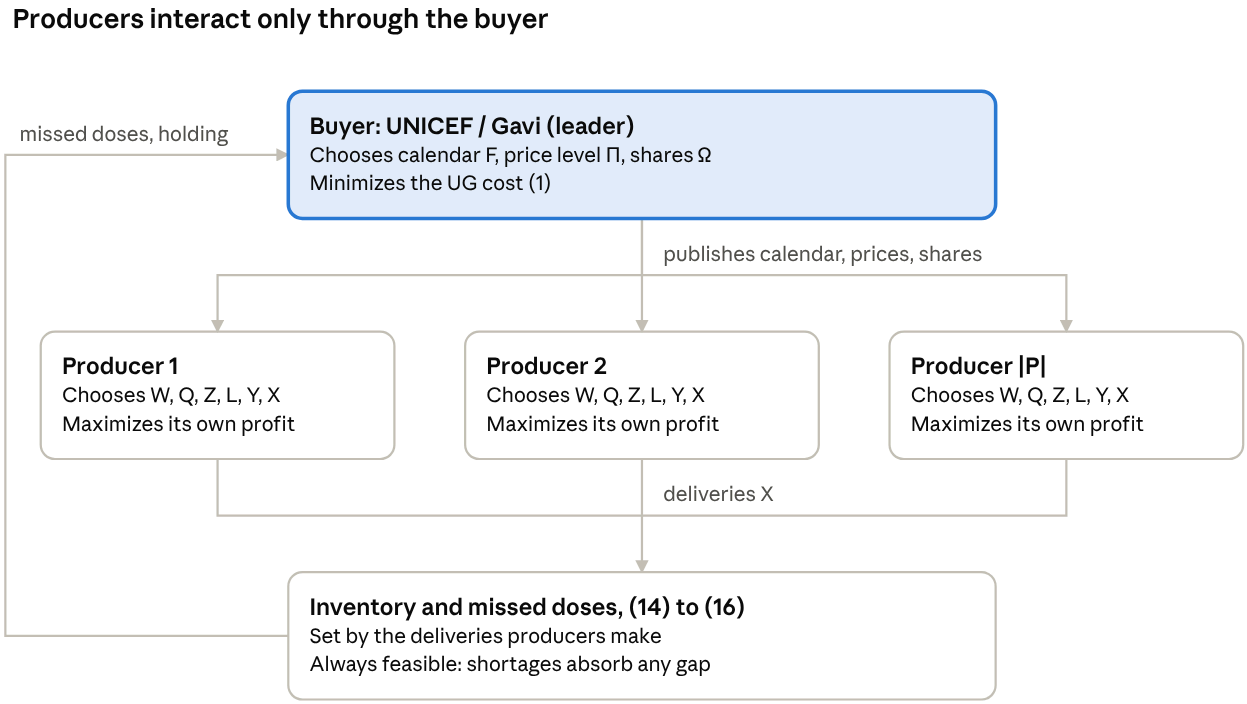
Summary

The buyer (UNICEF/Gavi) leads by choosing the tender calendar F, a per-dose price from a small ladder, and each producer's award share. Each producer then maximizes its own profit by choosing participation W, commitments Q, discount bracket Z, capacity expansion L, production Y and deliveries X. Almost every constraint comes straight from the manuscript; the changes are:

- The UG objective (1) becomes the leader's objective, with the fixed price replaced by the leader's price.
- The MP objective (20) becomes each producer's objective, split per producer, plus a synthetic per-dose production cost.
- The ROI constraint (8) is dropped: participation is now a best response, and staying out (all zeros) is always feasible, so no producer accepts a loss.
- Constraint (12) becomes an equality on tender-indexed deliveries, so a producer cannot be paid for doses it never delivers.
- Constraints (2) and (6), which made the buyer guarantee a producer would show up, are dropped from the leader. A tender nobody joins now shows up as missed doses, penalized by β.

Because producers interact only through the buyer, the lower level splits into one small MILP per producer. That keeps the model solvable with MibS on small instances and with a value-function decomposition at larger scale.

Structure



game structure · one leader, independent producer followers

The buyer moves first and commits to the calendar and terms. Each producer responds on its own, and the buyer's cost depends on what the producers deliver.

Notation

Everything not listed here keeps its manuscript meaning (A, V, P, T, 𝒟, A_p, V_p, P_a, P_v, A_v, δ_t, g_t, β, h_v, r̄_vt, r_vpt, ζ_vm, ϕ^L_vm, ϕ^U_vm, s̄_p, s_pt, κ, γ_p, f_vptτ, d_at, Î_v0, Δ).

Symbol	Type	Meaning	Status
K, ρ_vtk	set, parameter	Price ladder levels k for vaccine v, e.g. 0.9, 1.0, 1.1 × r̄_vt	New
J, υ_j	set, parameter	Award-share levels, e.g. 0, 0.5, 1	New
d̰_vt	parameter	Vaccine-level demand forecast (the Linksbridge input before antigen aggregation)	Existing data, new role

Symbol	Type	Meaning	Status
D̂_vtτ	parameter	Tender volume for v over [t, τ]: the sum of d̂_vl for l from t to τ	New, derived
c_vp	parameter	Synthetic per-dose production cost	New
F_atτ	binary, leader	Tender for antigen a over [t, τ]	Existing
Π_vtk	binary, leader	Price level k chosen for vaccine v in tenders starting at t	New
Ω_vpttj	binary, leader	Award-share level j for producer p, vaccine v, tender [t, τ]	New
I_vt, V_vt, S_at	continuous, leader	Inventory, doses used, missed doses, driven by deliveries	Existing
W_ptτ	binary, follower	Producer p joins tender [t, τ]	Existing, now a best response
Q_vptτm	continuous, follower	Commitment in discount bracket m	Existing
Z_vptm	binary, follower	Active discount bracket	Existing
L_pt	integer, follower	Capacity expansion	Existing
Y_pt	binary, follower	Producing in period t	Existing
X_vpttl	continuous, follower	Delivery in period l against tender [t, τ]	New index on X
Γ_vpttmk	continuous, follower	Commitment Q_vptτm sold at price level k (exact split of Q by the chosen level)	New, auxiliary

The manuscript's per-period delivery is recovered as X_vpl = the sum of X_vpttl over all tenders [t, τ] that contain l.

Upper level: the buyer

The buyer minimizes the UG cost (1), with procurement now priced at the ladder level it chose. Γ carries the product of price choice and commitment (see Linearization).

$$\min_{F,\Pi,\Omega,I,V,S}\ \sum_{t\in T}\delta_t\Bigl(\sum_{a\in A}\sum_{(t,\tau)\in\mathcal F} g_t F_{at\tau} +\sum_{v\in V}\sum_{p\in P_v}\sum_{\tau}\sum_{m\in M}\sum_{k\in K}\rho_{vtk}(1-\zeta_{vm})\,\Gamma_{vpt\tau mk} +\sum_{a\in A}\beta S_{at}+\sum_{v\in V}h_v\bar r_{vt}I_{vt}\Bigr)$$

Subject to:

$$\begin{aligned}
&\text{(3), (4)} && \text{at most one start per period; rolling coverage}\\
&\textstyle\sum_{k\in K}\Pi_{vtk}\le \sum_{a\in A_v}\sum_{\tau:(t,\tau)\in\mathcal F}F_{at\tau} && \forall\, v,t \quad \text{price only when a tender opens}\\
&\textstyle\sum_{k\in K}\Pi_{vtk}\le 1 && \forall\, v,t\\
&\textstyle\sum_{j\in J}\Omega_{vpt\tau j}\le \sum_{a\in A_v}F_{at\tau} && \forall\, v,\ p\in P_v,\ (t,\tau)\in\mathcal F\\
&\textstyle\sum_{j\in J}\Omega_{vpt\tau j}\le 1 && \forall\, v,\ p\in P_v,\ (t,\tau)\in\mathcal F \quad \text{one share level}\\
&I_{v,t-1}+\textstyle\sum_{p\in P_v}\sum_{(t',\tau)\ni t}X_{vpt'\tau t}=V_{vt}+I_{vt} && \forall\, v,t \quad \text{(14), deliveries from followers}\\
&\text{(15), (16)} && \text{missed doses, initial inventory}\\
&\Gamma\text{ is defined in the follower} && \text{price split of }Q\text{, see Linearization}\\
&(W_p,Q_p,Z_p,L_p,Y_p,X_p)\in\mathcal R_p(F,\Pi,\Omega) && \forall\, p\in P\\
&F,\Pi,\Omega\in\{0,1\};\quad I,V,S\ge 0
\end{aligned}$$

The inventory and missed-dose constraints contain follower deliveries, but they can never be violated: the buyer's V, I and S always adjust to whatever is delivered. So the only real coupling between levels runs through the objective, which keeps the model well posed.

Lower level: each producer

Given the buyer's calendar, prices and award shares, producer p maximizes the MP profit (20) for its own vaccines, less a per-dose production cost. The buyer's price choice enters only through the bounds on Γ (see Linearization), so each producer's problem is an ordinary MILP with constant objective coefficients.

$$\mathcal R_p(F,\Pi,\Omega)=\arg\max_{W,Q,Z,L,Y,X}\ \sum_{t\in T}\delta_t\Bigl(\sum_{v\in V_p}\sum_{\tau}\sum_{m\in M}\sum_{k\in K}\rho_{vtk}(1-\zeta_{vm})\,\Gamma_{vpt\tau mk} -\gamma_p L_{pt}-\sum_{v\in V_p}\sum_{\tau}f_{vpt\tau}W_{pt\tau} -\sum_{v\in V_p}\sum_{(t',\tau)\ni t}c_{vp}X_{vpt'\tau t}\Bigr)$$

Subject to:

$$\begin{aligned}
&W_{pt\tau}\le \textstyle\sum_{a\in A_p}F_{at\tau} && \forall (t,\tau)\in\mathcal F \quad \text{(5), F is now data}\\
&\textstyle\sum_{m}Q_{vpt\tau m}\le \hat D_{vt\tau}\sum_{j\in J}u_j\,\Omega_{vpt\tau j} && \forall v\in V_p,(t,\tau) \quad \text{award ceiling}\\
&\textstyle\sum_{m}Q_{vpt\tau m}\le \hat D_{vt\tau}\,W_{pt\tau} && \forall v\in V_p,(t,\tau) \quad \text{commit only if joined}\\
&\textstyle\sum_{l=t}^{\tau}X_{vpt\tau l}=\sum_{m}Q_{vpt\tau m} && \forall v\in V_p,(t,\tau) \quad \text{(12) as equality}\\
&\textstyle\sum_{v\in V_p}\sum_{(t',\tau)\ni t}X_{vpt'\tau t}\le Y_{pt}\bigl(s_{pt}+\sum_{l\le t}\kappa\bar s_pL_{pl}\bigr) && \forall t \quad \text{(13)}\\
&\text{(7), (9)}\text{--}\text{(11), price split of }Q && \text{commitment capacity, discount brackets, }\Gamma\\
&W,Z,Y\in\{0,1\},\ L\in\mathbb Z_+,\ Q,X\ge 0
\end{aligned}$$

The McCormick linearizations of (7) and (13) from Appendix B carry over unchanged, since both products stay inside the follower. Setting every variable to zero is always feasible and earns zero, so no producer ever accepts a tender at a loss; that is what replaces the ROI constraint (8).

Linearization and modeling choices

Price times quantity, made linear and solver-ready. Exact MIBLP solvers such as MibS need the producer's objective to have constant coefficients, so the price cannot multiply Q directly. Instead, each commitment is split across price levels by a follower variable Γ, and the buyer's choice of level switches the split on:

$$\sum_{k\in K}\Gamma_{vpt\tau mk}=Q_{vpt\tau m},\qquad 0\le\Gamma_{vpt\tau mk}\le \hat D_{vt\tau}\,\Pi_{vtk}$$

Because at most one Π_vtk equals 1, all of Q sits at the chosen level and no McCormick bounds are needed. Both objectives then use ρ_vtk (1 − ζ_vm) Γ_vptτmk, with ρ_vtk = λ_k r̄_vt for ladder multipliers λ_k such as 0.9, 1.0 and 1.1.

Pay on commitment, enforce delivery. With (12) as an inequality, a profit-maximizing producer would commit, collect revenue and never deliver. The equality, on deliveries indexed by tender, closes that loophole. It also fixes a double-count in the manuscript's (12) when one producer holds overlapping tenders for the same vaccine.

Discount brackets become a menu. ζ_vm and the bracket bounds stay parameters set by the buyer in advance, and each producer self-selects its bracket through Z. Making the bracket design a leader decision is a later extension.

Optimistic convention. When a producer is indifferent between two responses, the model assumes it picks the one better for the buyer. This is the standard choice and what MibS solves. To avoid relying on ties, set the lowest ladder level a small margin above each producer's cost; a pessimistic re-solve is a robustness check for the paper, not the first build.

High-point relaxation and the cost of autonomy. Dropping the producers' optimality but keeping their constraints gives one MILP: your UG model without (8), with the price ladder. Its optimum is a lower bound on the bilevel cost, and the difference is the cost of producer autonomy. Report UG with (8) alongside it, since (8) was the manuscript's approximation of producer rationality.

Synthetic producer data

Only one producer parameter is new: the per-dose production cost c_vp. Setup costs f_vptτ, expansion costs γ_p and capacities s_pt already exist in your data. Generate c_vp as a fraction of the reference price, so it stays consistent with the prices you already use:

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c_{vp}=\theta_p\,\bar r_{v},\qquad \theta_p\sim
U[\theta^{lo}_{g(p)},\theta^{hi}_{g(p)}]
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Here g(p) is the producer's size group (small, medium, large by base capacity s̄_p, as in the manuscript), and r̄_v is the average reference price over the horizon. The ranges below are placeholders to tune, not estimates:

Size group	θ range (placeholder)	Rationale
Large	0.40 to 0.60	Scale economies
Medium	0.50 to 0.70	
Small	0.60 to 0.80	Higher unit cost, the group most likely to stay out

Draw θ once per replication and reuse your 30-replication design, so every result is reported as a range across cost draws. One sanity check per instance: for each vaccine, at least one producer must earn a positive margin at the top ladder price after the deepest discount, or nobody will ever bid.

Solution approach and first instance

Build in this order. Each step reuses the one before and gives a usable result on its own.

1. **High-point relaxation in your existing Gurobi code.** Remove (8), (2) and (6) from the leader; add Π, Ω, Γ, the tender-indexed deliveries and c_vp. Solving it gives the lower bound and a candidate calendar.
2. **Best-response check.** Fix the candidate's F, Π and Ω, then solve each producer's MILP separately. Plugging the responses back into the buyer's cost gives a feasible bilevel solution, so steps 1 and 2 together already bound the cost of producer autonomy.
3. **Exact solve with MibS on the small instance.** Because producers never share a constraint, maximizing the sum of their profits gives the same responses as maximizing each one separately. So the lower level can go to MibS as a single follower. Export the model as MPS plus MibS's auxiliary file listing follower variables, constraints and objective.
4. **Value-function decomposition for larger instances** (Zeng & An 2014; Yue, Gao, Zeng & You 2019). The master is the high-point relaxation plus, for each producer, cuts built from best responses found so far. Each iteration solves the producer MILPs in parallel at the new candidate. All linking variables (F, Π, Ω) are binary, which these methods require.

First test instance. Measles-containing vaccines MMR, MR and M (three antigens: measles, mumps, rubella), the producers that supply them, T = 5 years, Δ = 3, three price levels (λ = 0.9, 1.0, 1.1), award shares {0, 0.5, 1}, and two discount brackets. Solve it with steps 1 to 3 before scaling to the five manuscript vaccine groups and the ten-year horizon.

The nesting (M inside MR inside MMR) is a small version of the Penta/Hexa overlap, so it tests the multi-antigen coverage logic early. Measles data would need adding, since the manuscript's instances cover only Hexa, Penta, Rotavirus, HPV and PCV.

Open decisions

These are calls I made to get a complete draft; each is easy to change.

Decision	Draft choice	Alternative
Delivery index	Tender-indexed X_vptτl with an equality to Q	Keep the manuscript's X_vpt and pay on delivery instead of commitment
Award shares	Leader picks a share level per producer, which keeps producers independent	Drop shares and let producers compete for one volume (an equilibrium problem, much harder)
Coverage guarantees (2), (6)	Removed from the leader; empty tenders show up as missed doses	Keep them as coupling constraints, which MibS can handle but which can make the model infeasible for some calendars
Price ladder	Three levels at 0.9, 1.0, 1.1 × r̄_vt	More levels, or levels tied to each producer's reservation price r_vpt
Cost ranges	Placeholder θ ranges by size group	Ranges tied to published price spreads across producers